

are listed below.

- **Thread group:** This is a collection of threads, and each thread represents one user accessing the application in it. Ideally, it simulates one real user making one request to the application server. We can have multiple threads configured here.
- **Sampler:** This is the type of request, such as FTP, HTTP, etc.
- **Listener:** This shows the results in different formats.
- **Configuration:** Sets up defaults and variables for later use by samplers.
- **Assertion:** Validates whether a response from the server is expected or not.

Load testing

Prerequisites

First, we will test the Google home page.

- Open the installed JMeter.
- Click on *File -> New*.
- Update the name to *GoogleHomePageTestPlan*.
- Click on *Save Icon #*; it will be saved as a *.jmx* file.

Step 1: Thread group

In the thread group, we need to know the following.

- **Number of threads:** 'N' number of users connecting to the target URL.
- **Ramp-up period:** This allows to include delay when requests hit the target. As an example, if we have 10 threads and 5 as the ramp-up period, the 10 requests should hit the target within 5 seconds with 0.5 second delay between each request. This is primarily used when we don't want to overburden the server as well as the client. But if you want to do stress testing, try tweaking the property, based on the use case.
- **Loop count:** This is 'N' number of times the test case is to be executed; for example, if you have 5 threads and the loop count is 2, the request will hit the target 10 times.

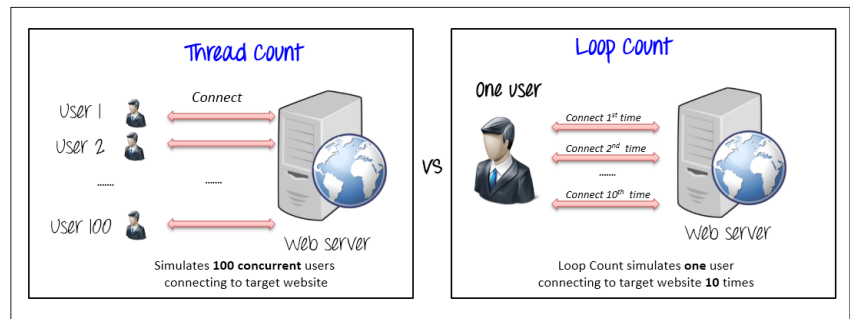


Figure 2: Difference between thread count and loop count (Source: <https://medium.com/skysdigital/performance-testing-with-jmeter-for-dummies>)

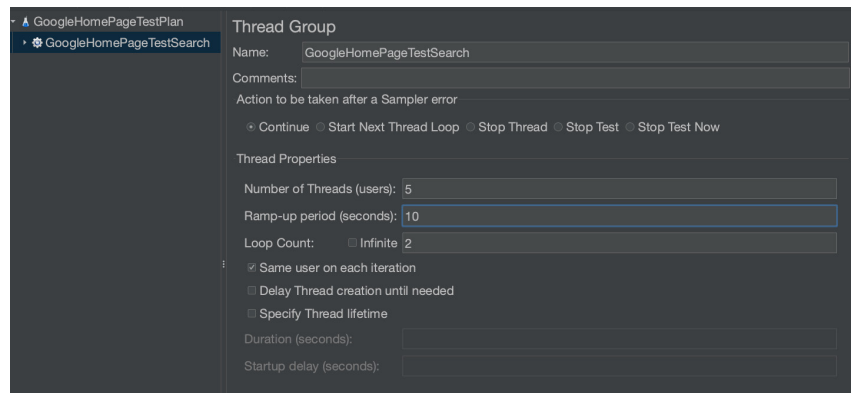


Figure 3: Thread group creation page

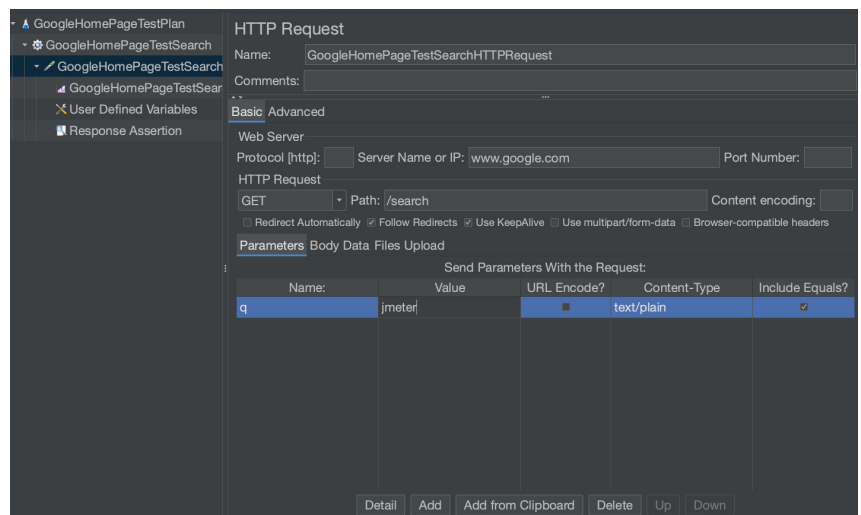


Figure 4: HTTP sampler creation page

- Now, let's carry out the steps.
- Right click *GoogleHomePageTestPlan -> Add -> Threads (Users) -> Thread Group*.
- It is a good practice to change the name, as readability matters.

For this, update the name to *GoogleHomePageTestSearch*.

- Update *Number of Threads (users)* to 5.

Note: Please do not update number of threads to more than 10. You may be blocked.